

Project Design Phase
Proposed Solution Template

Date	11 July 2026
Team ID	
Project Name	Electrical Vehicle Performance & Market Analysis Dashboard
Maximum Marks	2 Marks

Proposed Solution Template

Project owner shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1	Problem Statement (Problem to be Solved)	The rapid growth of electric vehicles has generated large amounts of data related to vehicle specifications, charging infrastructure, pricing, and performance. However, this information is scattered across multiple datasets, making it difficult for users, researchers, and businesses to analyze and compare EV models effectively. There is a need for a centralized analytics platform that provides meaningful insights through interactive visualizations.
2	Idea / Solution Description	The proposed solution integrates multiple electric vehicle datasets into a single MySQL database after performing data cleaning and preprocessing using SQL. The cleaned data is visualized using Tableau Public to build interactive dashboards and stories that analyze EV performance, top speed, driving range, acceleration, charging infrastructure, pricing, and regional charging station distribution. Interactive filters enable users to explore insights based on brand, body style, region, and price range.
3	Novelty / Uniqueness	Unlike static reports, this solution combines multiple EV datasets into one unified analytical platform. The Tableau Dashboard and Story provide interactive exploration of EV performance and charging infrastructure, enabling users to compare different brands and models visually. The integration of SQL-based data cleaning with Tableau storytelling offers an intuitive decision-support system for EV analysis.
4	Social Impact / Customer Satisfaction	The dashboard helps customers choose suitable electric vehicles based on performance, range, speed, and price. It also supports researchers, automobile manufacturers, dealerships, and policymakers by providing insights into charging infrastructure and market trends. Better data-driven decisions can encourage EV adoption and contribute to sustainable transportation.

S.No	Parameter	Description
5	Business Model (Revenue Model)	The solution can be offered as a subscription-based EV analytics platform for automobile companies, dealerships, EV startups, and research organizations. Revenue can also be generated through customized dashboards, consulting services, premium reports, and enterprise analytics solutions for organizations working in the electric mobility sector.
6	Scalability of the Solution	The solution can be expanded by integrating live EV sales data, real-time charging station APIs, battery health information, traffic conditions, IoT sensor data, and government EV datasets. The platform can scale to support city-level, state-level, and national-level EV analytics without major changes to the existing architecture.